LAB 2: Variables, Constants, Operators, & Control Flow

- **Syllabus Alignment:** Unit 1: Introduction (10 LHs) — Variables, Constants, Data Types, Operators, Control Statements
- **Lab Objectives:** To manage dynamic variables, initialize immutable values, apply operators, and route data using conditional branches and iterative loops.
- **Lab Questions:**
    1. **Primitives & Math Operations:** Write a script that declares variables for tracking numeric primitives (e.g., product counts, prices). Calculate their arithmetic sum and product, and display the results cleanly.
    2. **Constants vs. Variables:** Define a global configuration constant (e.g., `COLLEGE` or `CURRENCY`) using the `define()` utility. Write a text block showing how it behaves differently from a standard, mutable variable instance.
    3. **Operator Matrix Evaluation:** Create a program that demonstrates the practical use of multiple operator categories: Arithmetic (`+`, `-`, `*`, `/`, `%`), Assignment (`+=`, `-=`), Logical combinations (`&&`, `||`, `!`), and Pre/Post Increment structures (`$i++`, `++$i`).
    4. **Conditional & Iterative Statements:** Write an application that uses an `if-else` block to evaluate numeric marks (Pass/Fail matrix). Follow it with a `for` loop that prints a sequential list of integers from 1 to 5.
    5. **Alternative Control Loop Flow:** Implement a script utilizing `while` and `do-while` variations to scan through a collection, using `break` and `continue` keywords to modify the loop execution paths dynamically.